Assignment 1 Sequence Bioinformatics

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1 Global local overlap alignment

Global alignment output:

Align_NinaRopertz global match=1 mismatch=-1 gap=1

X: X (8 letters)

Y: Y (5 letters)

Score: 2 (0 ms)

CTAATGCG

-T-ATG-G

Local alignment output:

Align_NinaRopertz local match=1 mismatch=-1 gap=1

X: X (8 letters)

Y: Y (5 letters)

Score: 3 (0 ms)

ATG

ATG

Overlap alignment output:

Align_NinaRopertz overlap match=1 mismatch=-1 gap=2

X: X (7 letters)

Y: Y (6 letters)

Score: 4 (0 ms)

TATT

TATT

Exercise 1.4 with global alignment output:

Align_NinaRopertz global match=2 mismatch=-1 gap=1

X: X (5 letters)

Y: Y (5 letters)

Score: 7   (1 ms)

CAGTG

CATTG

2 Two mitochondrial sequences

Small.fasta

global:

Align_NinaRopertz global match=1 mismatch=-1 gap=1

X: human (247 letters)

Y: mouse (207 letters)

Score: 65   (8 ms)

Local:

Align_NinaRopertz local match=1 mismatch=-1 gap=1

X: human (247 letters)

Y: mouse (207 letters)

Score: 84   (10 ms)

Overlap:

Align_NinaRopertz overlap match=1 mismatch=-1 gap=1

X: human (247 letters)

Y: mouse (207 letters)

Score: 83   (7 ms)

Mtdna-200

Global:

Align_NinaRopertz global match=1 mismatch=-1 gap=1

X: NC_012920.1 Homo sapiens mitochondrion, complete genome (2000 letters)

Y: NC_005089.1 Mus musculus mitochondrion, complete genome (2000 letters)

Score: 193 (44 ms)

Local:

Align_NinaRopertz local match=1 mismatch=-1 gap=1

X: NC_012920.1 Homo sapiens mitochondrion, complete genome (2000 letters)

Y: NC_005089.1 Mus musculus mitochondrion, complete genome (2000 letters)

Score: 771 (57 ms)

Overlap:

Align_NinaRopertz overlap match=1 mismatch=-1 gap=1

X: NC_012920.1 Homo sapiens mitochondrion, complete genome (2000 letters)

Y: NC_005089.1 Mus musculus mitochondrion, complete genome (2000 letters)

Score: 771 (45 ms)

- ➔ To compare two homologous genomes the global alignment is the appropriate one. The human and mouse genome are related in length and gene order over the complete length of the genome. The local alignment would only present the most similar regions while the overlap alignment would only be suitable if the two genomes are overlapping at the start or the end of one sequence and the other.
- ➔ To aligne the complete genome we would need a matrix of size 16,500 x 16,500. This would equal 270 million matrix-cells. It would therefore require a huge memory because time and memory grow with n*m, so it does not scale to larger genomes.

3 Which table is which?

A = overlap alignment

-> first row and column initialized with 0 but also negative matrix entries

optimal score = 3

cells visited: [4][6], [3][5], [2][4], [1][3], [0][2]

Alignment:

TTAG

TTAG

B = global alignment

-> first row and column initialized with -i*d and -j*d

optimal score = 1

cells visited: [5][6], [4][5], [3][5], [2][4], [1][3], [0][2], [0][1], [0][0]

Alignment:

CATTAG -

- - TTAGG

C = local alignment

-> no score < 0

-> optimal score = 4

-> cells visited: [4][6], [3][5], [2][4], [1][3], [0][2]

Alignment:

TTAG

TTAG

4 All optimal alignments

There are two optimal alignments which part at cell [2][4]

CTAATGCG

- T–ATG -G

And

CTAATGCG

- TA - TG-G

5 Negative expected score

a)

if all four bases are equally frequent we can expect a random match with a probability of ¼ and a mismatch with ¾ therefore is the expected value:

E = M*1/4 + -1*3/4

E = (-3 + M) / 4

If we use Integers as scores for M = 1, M = 2 and M = 3 the expected score of a random pair would not be positive

…

b)

possible pairings:

p(LL) = 4 * (0.5*0.5) = 1

p(LK) = -2 * (0.5*0.3) = -0.3

p(KL) = -0.3

p(LF) = 0

p(FL) = 0

p(KK) = 5 * (0.3*0.3) = 0.45

p(KF) = -3 * (0.3*0.2) = -0.18

p(FK) = -0.18

p(FF) = 6 * (0.2*0.2) = 0.24

E = p(LL) + 2 * p(LK) + 2 * p(LF) + p(KK) + 2 * p(KF) + p(FF) = 1 + -0.6 + 0 + 0.45 + - 0.36 + 0.24 = 0.73

E = 0.73 > 0

It does not contradict the requirement of a negative score, as the three-residue matrix only depicts a small part of the actual BLOSSUM62 matrix. With the real amino acid frequencies of all amino acids that the matrix provides, the expected score is negative.

6 Reading an alignment program